# Ezhill Ragesh

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Frontend / Platform Engineer – Design Systems, Performance & Accessibility Tooling, AI-Assisted Engineering

## Technical Skills

**Languages**: TypeScript, JavaScript (ES6+), Python, SQL, C/C++
**Frontend & Design Systems**: React, Next.js, Storybook, Tailwind CSS, HTML5, CSS3/SCSS, Vite, Redux
**Platform & Tooling**: Chrome Extensions (MV3), Vitest, esbuild, Git, GitHub Actions, CI/CD
**Performance & Accessibility**: Core Web Vitals, WCAG/ARIA, axe-core, Lighthouse, Perf Profiling
**AI/LLM & Backend**: Agent-Assisted Engineering Workflows, Node.js, Express.js, REST APIs, PostgreSQL/MongoDB

## Professional Experience

**Software Engineer I** Feb. 2025 – Present
*Ramco Systems* *Chennai, India*

- **Own** reusable component development for **Nebula**, Ramco's org-wide React/TypeScript design system consumed by **ERP, HRM, Aviation, and Logistics** platforms; **migrated the build pipeline from Rollup to Vite**, cutting library size **89% (35MB → 4MB)**.
- **Led** the **SCSS → Tailwind/design-token** migration, building shared theming infrastructure across **light/dark/RXD** variants while eliminating cross-theme style leakage; **cut Storybook token CSS 88% (13MB → 1.6MB)**.
- **Improved accessibility across the component library** — ARIA semantics, focus management, and keyboard interactions across dialogs, popovers, tables, and forms — aligning with WCAG conformance targets validated through an independent third-party audit, without breaking existing component APIs.
- **Optimized rendering performance** across data-heavy product surfaces through **lazy loading, virtualization, and memoization**, measurably reducing initial load time and runtime memory pressure.
- **Designed an agent-assisted engineering workflow** for component changes – automated Storybook reproduction, visual QA, **backward-compatibility and blast-radius validation**, and PR review – reducing manual regression-testing effort across the design-system team.
- **Own release engineering** for the shared library – package publishing, build/declaration pipelines, and git-history root-cause analysis – so changes land safely across multiple consuming products.

## Projects

**Performance Engine** | *TypeScript, Chrome Extension (MV3), React Internals*

- **Architected** a real-time performance diagnostics engine that instruments **React fiber commits**, DOM/network events, and long tasks, correlating them into **causal interaction traces** to explain root cause – not just symptoms – inside a custom Chrome DevTools panel.
- **Built** an evidence-based analyzer (detection rules + a budgets engine) computing confidence via **noisy-OR scoring** over measured evidence; benchmarked at **55ms for 10k events** with linear scaling.

**SourceGuide** | *TypeScript, shadcn/ui, Tailwind CSS*

- **Developed** a platform surfacing beginner-friendly GSoC issues by integrating the **GitHub API** for filtering and enrichment; deployed on Vercel, growing to **1,000+ users** at **30 users/day**.

**TrueLead** | *TypeScript, Express.js, Gemini LLM*

- **Built** an AI-powered product-research platform that analyzes uploaded images and generates actionable insights via an **Express.js** backend and custom **Gemini LLM** prompt pipelines.

## Education

**B.Tech in Computer Science Engineering and Business Systems** Sept. 2021 – July 2025
*Vellore Institute of Technology, Vellore* *CGPA: 8.4*

## Volunteer Experience

**Technical Head** Apr. 2022 – Jan. 2024
*Zero Waste Management Club*

- Led sustainability-focused events and workshops, driving high participation and engagement.